SEMICONDUCTORS II

Trade War, Technology Policy, and the Global Semiconductor Market

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Preliminary Report - Phases 1 and 2

Data construction and validation; exploratory analysis and time-series diagnostics

Period analyzed: 2010-2025

Working paper - November 2026

Abstract

This report presents the preliminary results of the first two phases of Semiconductors II. Phase 1 focused on constructing, reconstructing, and validating the underlying datasets. Phase 2 examined the global and regional evolution of the market and the statistical properties of the series before introducing structural-break or policy-intervention models.

Between 2010 and 2025, the global semiconductor market exhibits strong long-run expansion, recurrent sectoral cycles, and pronounced seasonality. Regional rebalancing is uneven: America increases its market share, while Europe and Japan lose relative weight. From 2015 onward, the available decomposition shows strong absolute growth in China, although China's share within Asia Pacific peaks at 57.9% in September 2018 and ends 2025 at 47.7%.

Unit-root tests indicate that first log differences display substantially more favorable stationarity properties than level series. Serial dependence and seasonality nevertheless remain significant, implying that subsequent inference must model these components explicitly. The comparison between the original monthly series and the three-month moving average confirms that smoothing preserves long-run levels very closely but materially changes short-run volatility and autocorrelation.

These findings do not constitute causal evidence on tariffs, export controls, subsidies, or industrial policy. Their purpose is to establish the empirical basis and specification requirements for Phase 3, which will focus on structural breaks and policy-intervention models.

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1. Introduction and research question

Semiconductors II examines whether trade, industrial, and technology policies associated with the growing strategic rivalry between the United States and China were linked to statistically significant changes in global and regional semiconductor market dynamics.

1.1. Research hypothesis

The project's central substantive hypothesis is formulated as follows:

> **H0 (null hypothesis):** trade and technology policies associated with US-China rivalry did not produce statistically significant changes in semiconductor sales dynamics.
>
> **H1 (alternative hypothesis):** those policies were associated with statistically significant changes in the level, trend, growth, or regional distribution of semiconductor sales.

These hypotheses are not yet tested directly in Phases 1 and 2. The historical periodization is treated as a working hypothesis rather than an econometric conclusion: expected breaks derived from policy chronology will later be compared with breaks detected from the data. The tests conducted in Phase 2 evaluate prior statistical properties of the series - stationarity and autocorrelation - that are necessary to specify correctly the models that will test the central hypothesis in Phase 3.

The first two phases therefore address a prerequisite to causal identification: ensuring data quality and consistency, describing the main patterns, and determining which transformations and error structures will be required in subsequent models.

2. Phase 1. Data construction and validation

Phase 1 developed a reproducible acquisition, auditing, and validation pipeline using WSTS Historical Billings and monthly market reports published by SIA/WSTS. The WSTS historical dataset was processed both in its original monthly frequency and as a three-month moving average. After this first definition, the report refers to the latter as the “three-month moving average” or the “smoothed series”.

Table 1. Dataset construction and validation

Dataset	Period	Coverage	Missing	Main validation
WSTS monthly	2010-2025	192 months / 960 rows	0	Regional sum valid in 192/192 months
WSTS 3-month moving average	2010-2025	192 months / 960 rows	0	Regional sum valid in 192/192 months
SIA/WSTS regional smoothed series	2015-2025	132 months	0	120/120 PDFs parsed; max total discrepancy = USD 0.02 bn

The regional SIA/WSTS series was reconstructed from 120 monthly reports covering 2016-2025. All 120 documents were parsed successfully, and historical values included in the 2016 reports made it possible to reconstruct the twelve months of 2015. The resulting dataset contains a complete 132-month series for America, Europe, Japan, China, Asia Pacific / Other, and the worldwide total.

The comparison between values originally published in SIA/WSTS monthly reports and the currently available WSTS historical series shows correlations close to 1, although small level differences remain. To preserve internal consistency, the China-disaggregated dataset does not combine observations from different versions of the data or values subject to later revisions.

Throughout this report, “America” is the project's normalized label for the WSTS region named “Americas” in the source data. It must not be interpreted as the United States alone: the WSTS category aggregates the United States, Canada, Mexico, Brazil, and the rest of Latin America.

3. Empirical architecture

The project uses two complementary data layers. The first preserves the original monthly frequency and serves as the main basis for studying temporal dynamics, seasonality, and future structural breaks. The second allows China to be separated from the rest of Asia Pacific but is available as a three-month moving average.

Table 2. Empirical data architecture

Layer	Source	Period	Series	Main use
Layer 1	WSTS monthly	2010-2025	America, Europe, Japan, Asia Pacific, Worldwide	Temporal dynamics, seasonality, structural breaks
Layer 2	SIA/WSTS smoothed	2015-2025	America, Europe, Japan, China, Asia Pacific / Other, Total	China decomposition and relative analysis within Asia Pacific

The two layers are complementary, not interchangeable. Direct comparison shows that the moving average preserves long-run trajectories very closely while reducing volatility and changing the autocorrelation structure. This difference determines the role of each dataset in the following phases.

4. Phase 2. Exploratory analysis and time-series diagnostics

4.1. Monthly market evolution, 2010-2025

The original monthly series makes it possible to observe trend, cycles, and seasonality simultaneously. Worldwide semiconductor sales rise from approximately USD 22.1 billion in January 2010 to USD 85.3 billion in December 2025. This is a point-to-point comparison and should not be interpreted as a structural growth rate, since January and December occupy different positions within the seasonal pattern.

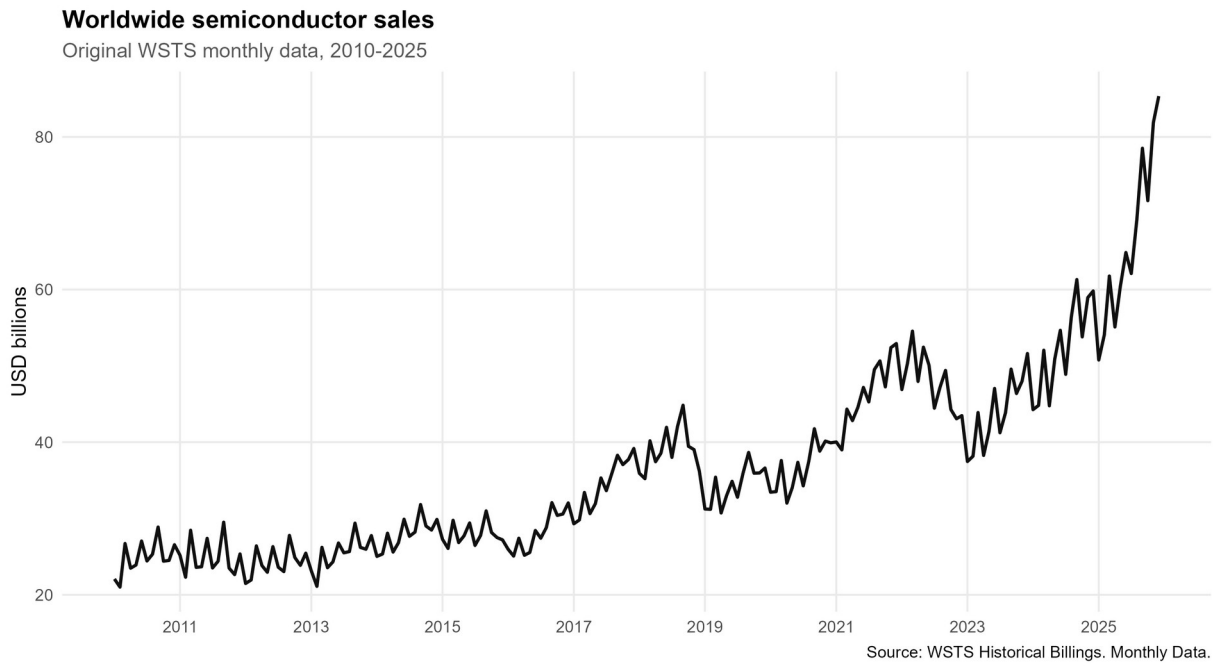


Figure 1. Worldwide monthly semiconductor sales, 2010-2025.
Source: author's calculations based on WSTS Historical Billings.

Growth is geographically uneven. America accelerates strongly in the second half of the sample, Asia Pacific remains the largest regional market, and Europe and Japan display substantially weaker relative growth.

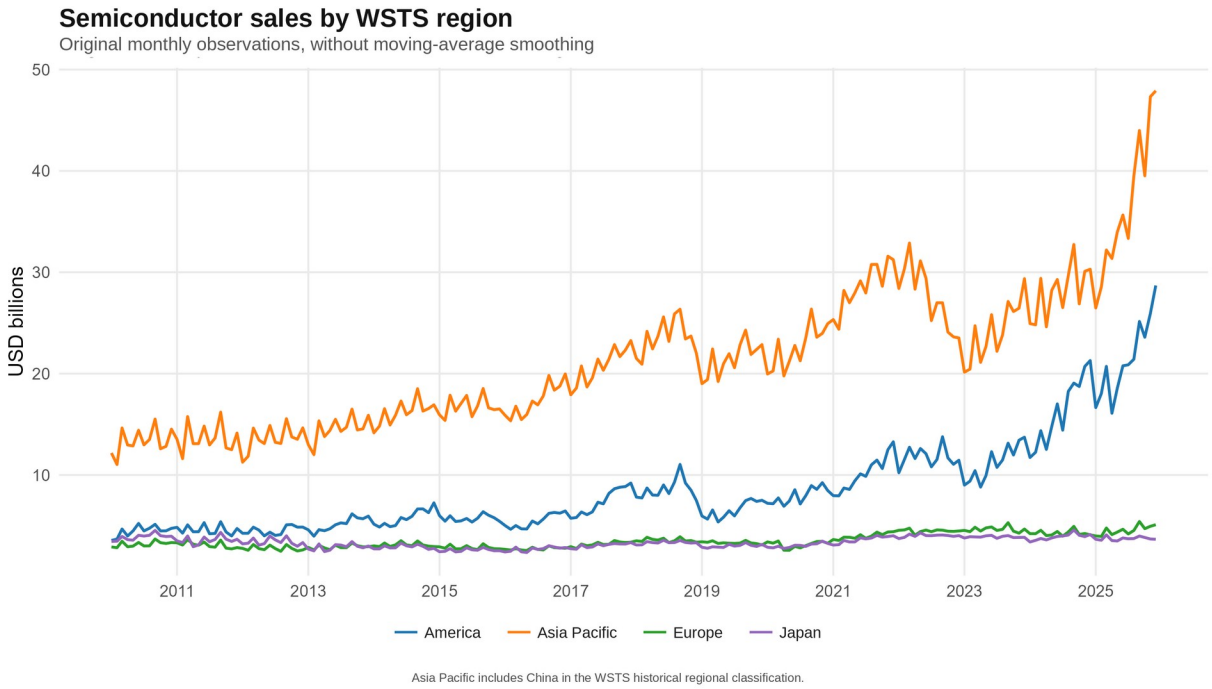


Figure 2. Monthly semiconductor sales by region, 2010-2025.
Source: author's calculations based on WSTS Historical Billings.

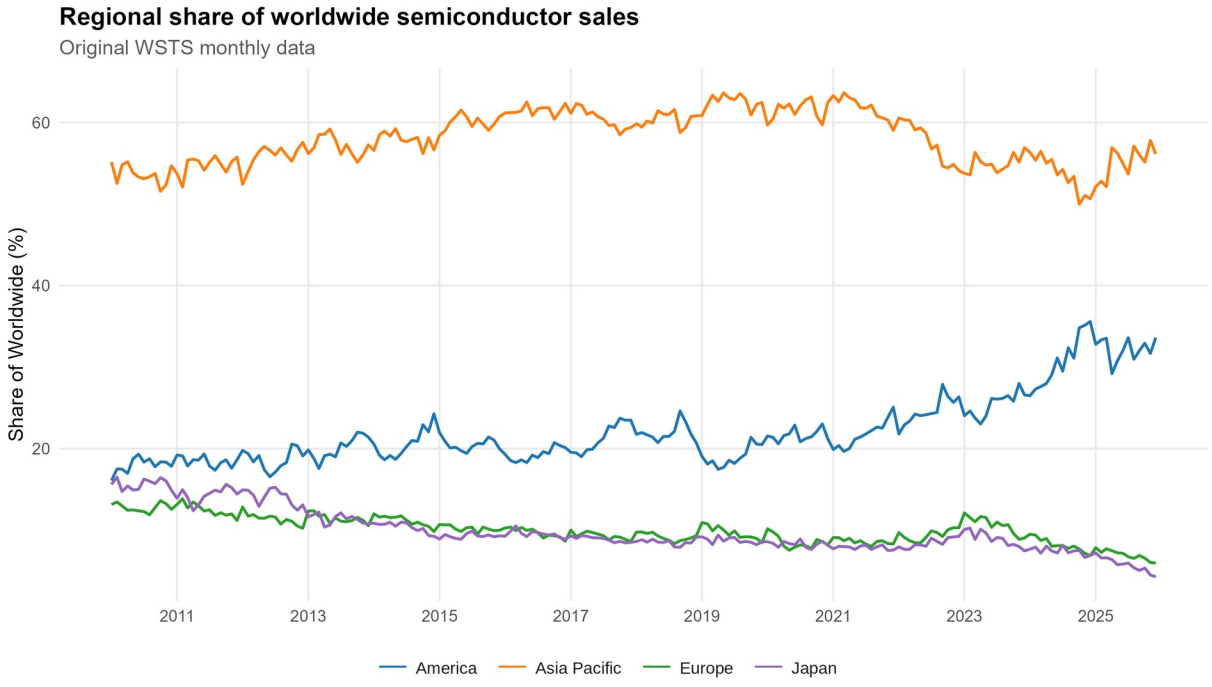


Figure 3. Regional shares of worldwide semiconductor sales, 2010-2025.
Source: author's calculations based on WSTS Historical Billings.

Year-over-year growth reveals pronounced sectoral cycles. The data show weakness around 2011-2012, expansion in 2017-2018, contraction in 2019, strong growth during 2020-2022, another correction in 2022-2023, and renewed acceleration in 2024-2025. These cycles are potential confounders for any later attribution to trade or technology policy.

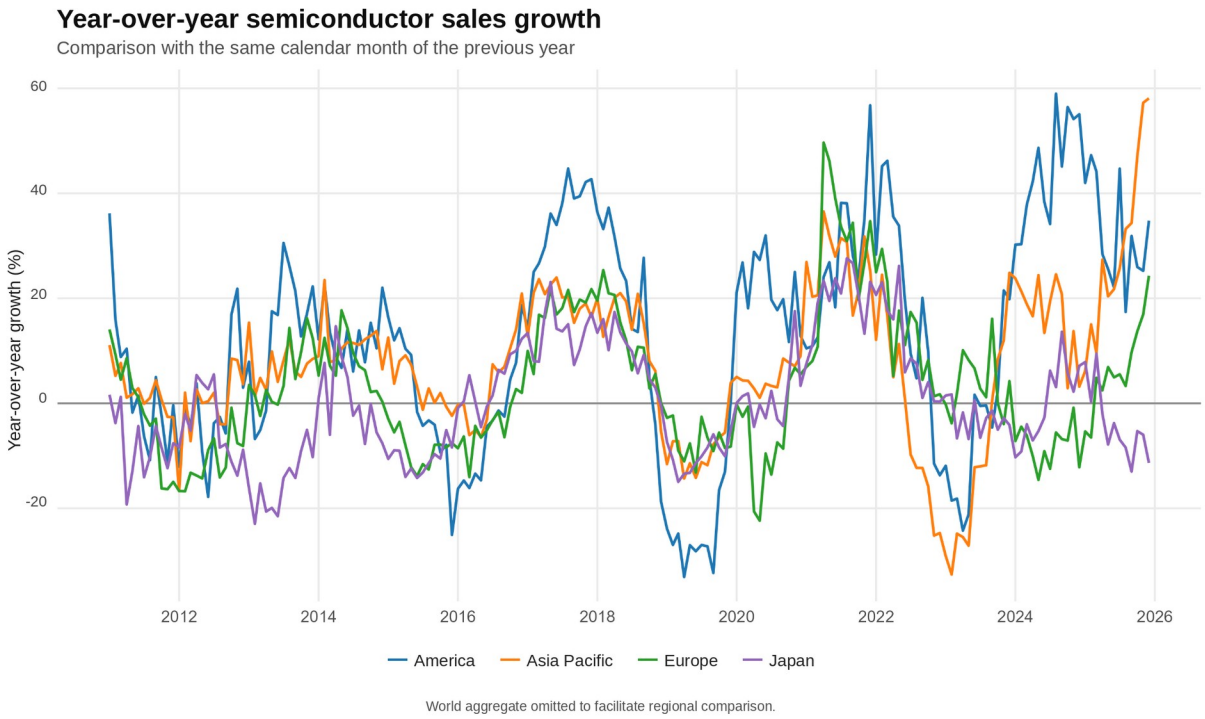


Figure 4. Year-over-year monthly semiconductor sales growth by region, 2011-2025.
Source: author's calculations based on WSTS Historical Billings.

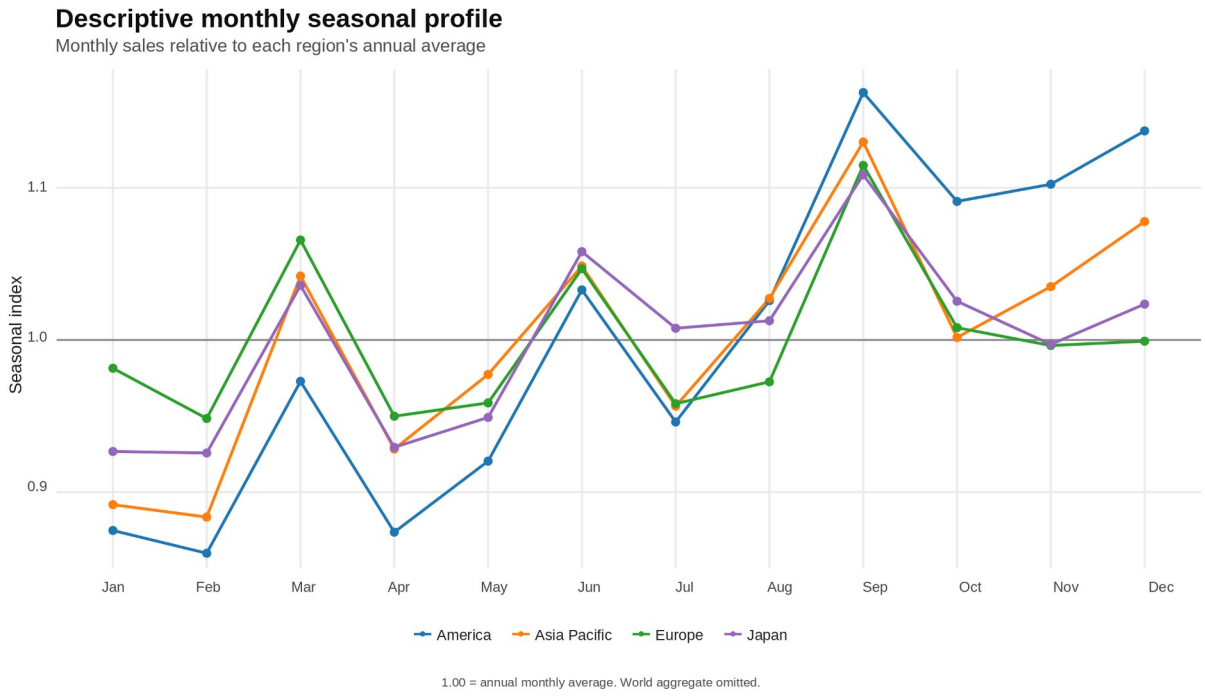


Figure 5. Descriptive monthly seasonal profile by region, 2010-2025.
Source: author's calculations based on WSTS Historical Billings.

4.2. China and the Asia Pacific decomposition, 2015-2025

The smoothed dataset allows China to be separated from the rest of Asia Pacific from 2015 onward. Between January 2015 and December 2025, America rises from USD 6.51 to 25.7 billion; China from 8.07 to 21.3; Asia Pacific / Other from 8.40 to 23.3; Europe from 2.95 to 4.86; Japan from 2.62 to 3.70; and the worldwide total from 28.6 to 78.9 billion.

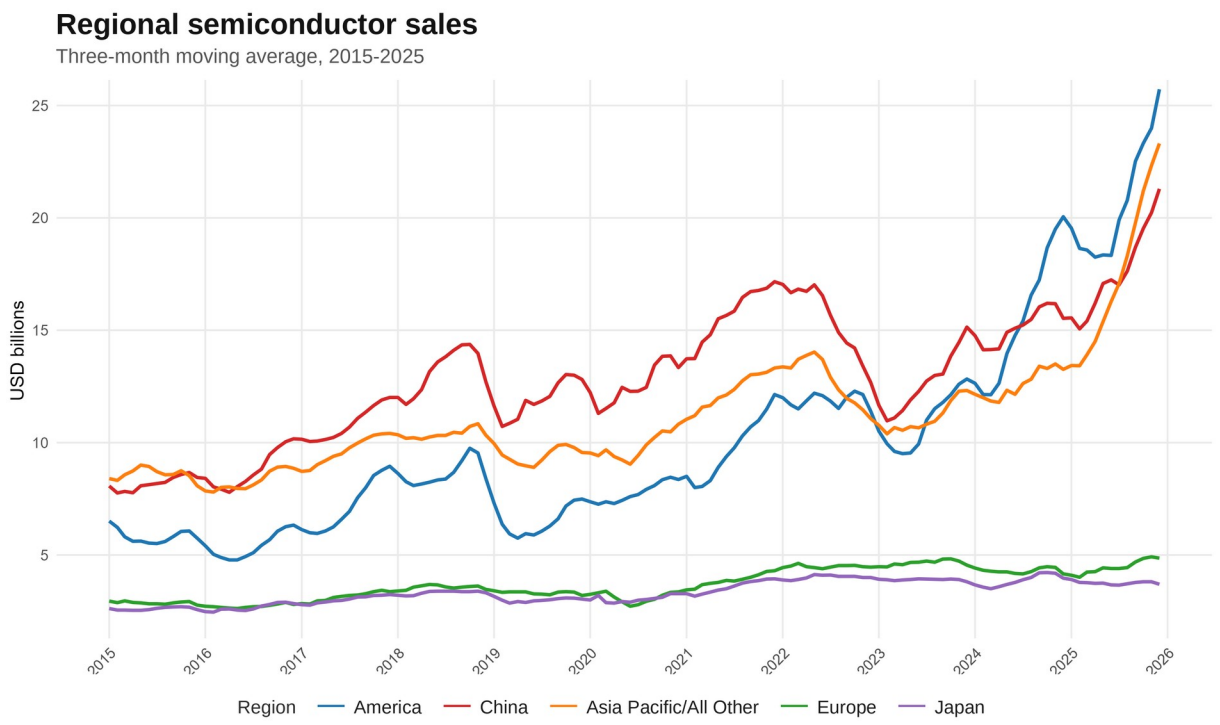


Figure 6. Regional semiconductor sales, smoothed using a three-month moving average, 2015-2025.
Source: author's calculations based on SIA/WSTS monthly market reports.

The reallocation of market shares is substantial: America gains approximately 9.8 percentage points between the beginning and end of the period, while Europe and Japan lose 4.17 and 4.49 percentage points, respectively. China ends the period with a slightly lower worldwide share than at the start despite strong absolute growth.

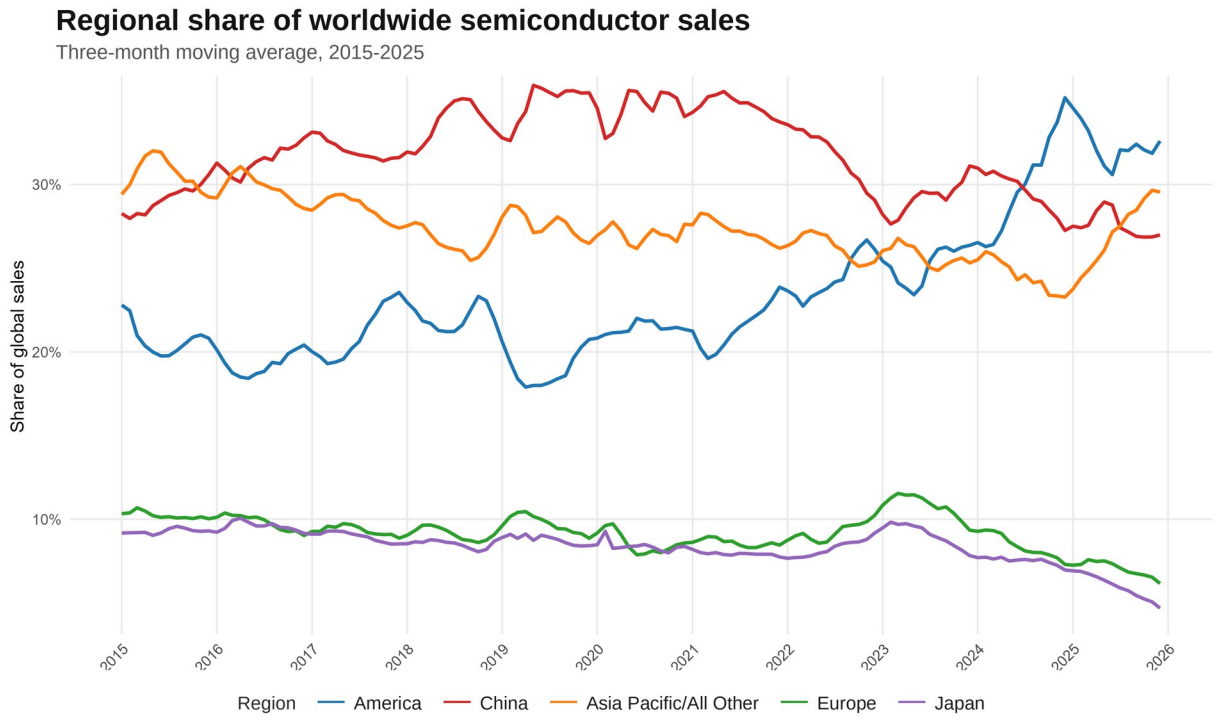


Figure 7. Regional shares of worldwide sales, smoothed series.
Source: author's calculations based on SIA/WSTS monthly market reports.

Within Asia Pacific, China's share starts at 49.0% in January 2015, reaches a maximum of 57.9% in September 2018, and declines to 47.7% in December 2025. This pattern motivates subsequent structural-break testing, but by itself does not identify a causal effect of the trade war.



Figure 8. China's share within Asia Pacific, 2015-2025.

Source: author's calculations based on SIA/WSTS monthly market reports.

4.3. Hypothesis tests on time-series properties

The diagnostics in this section are presented as formal hypothesis tests. A significance level of alpha = 0.05 is used. When p <= 0.05, the null hypothesis of the corresponding test is rejected; when p > 0.05, the evidence is insufficient to reject it. Econometrically, these tests help determine which transformations are appropriate and which dynamic structures must be incorporated before estimating structural-break or causal models.

4.3.1. Stationarity

For the Augmented Dickey-Fuller (ADF) and Phillips-Perron (PP) tests, the null hypothesis is H0: the series contains a unit root and is therefore non-stationary. The alternative is H1: the series is stationary. KPSS reverses the logic: H0 states stationarity around a level or trend, whereas H1 indicates non-stationarity. Reading tests with opposite null hypotheses together reduces the risk of drawing conclusions from a single statistic.

Table 3. Stationarity tests for the smoothed series

Series	Transformation	ADF p	KPSS p	Interpretation
America	Level	0.99	0.01	Strong evidence of non-stationarity
America	Log difference	0.01	0.1	Consistent with stationarity
Europe	Level	0.229	0.0629	Ambiguous result / possible trend stationarity
Europe	Log difference	0.0201	0.1	Consistent with stationarity
Japan	Level	0.539	0.0473	Evidence of non-stationarity
Japan	Log difference	0.01	0.1	Consistent with stationarity
China	Level	0.367	0.0368	Evidence of non-stationarity
China	Log difference	0.01	0.1	Consistent with stationarity
Asia Pacific / Other	Level	0.939	0.0257	Evidence of non-stationarity

Series	Transformation	ADF p	KPSS p	Interpretation
Asia Pacific / Other	Log difference	0.0923	0.0969	Ambiguous: ADF does not reject at 5%; KPSS does not reject stationarity
Total	Level	0.967	0.0247	Evidence of non-stationarity
Total	Log difference	0.01	0.1	Consistent with stationarity
China share of Asia Pacific	Level	0.854	0.01	Evidence of non-stationarity
China share of Asia Pacific	First difference	0.01	0.0757	Consistent with stationarity

Table 4. Stationarity tests for the monthly series

Series	Transformation	ADF p	KPSS p	PP p	Interpretation
America	Level	0.99	0.01	0.99	Strong evidence of non-stationarity
America	Log difference	0.01	0.1	0.01	Consistent with stationarity
Europe	Level	0.692	0.01	0.01	Mixed tests; do not infer I(1) from these tests alone
Europe	Log difference	0.01	0.1	0.01	Consistent with stationarity
Japan	Level	0.51	0.01	0.01	Mixed tests; problematic level series
Japan	Log difference	0.01	0.1	0.01	Consistent with stationarity
Asia Pacific	Level	0.962	0.0803	0.01	Mixed test results
Asia Pacific	Log difference	0.01	0.1	0.01	Consistent with stationarity
Worldwide	Level	0.99	0.01	0.0977	Substantial evidence of non-stationarity
Worldwide	Log difference	0.01	0.1	0.01	Consistent with stationarity

Test decision. For first log differences in the monthly dataset, ADF and Phillips-Perron reject the unit-root null for America, Europe, Japan, Asia Pacific, and Worldwide, while KPSS does not reject its stationarity null. Taken together, the evidence is consistent with stationarity of Δlog(Yt), supporting its use as a representation of monthly growth in part of the dynamic analysis.

The report does not yet conclude that every level series is formally I(1). Structural breaks can distort conventional unit-root tests and will be examined explicitly in the next phase.

4.3.2. Autocorrelation: Ljung-Box test

The Ljung-Box test evaluates H0: rho1 = rho2 = ... = rhok = 0, meaning no joint autocorrelation up to lag k. The alternative H1 states that at least one of these autocorrelations differs from zero. Lags of 6 and 12 months are among those examined.

Transforming the series does not eliminate temporal dependence. In the smoothed series, Ljung-Box strongly rejects the no-autocorrelation null. Part of this persistence is mechanical because adjacent three-month moving averages share two of their three monthly observations. However, the original monthly dataset also displays a pronounced temporal structure.

Table 5. Ljung-Box test for the smoothed series

Series	Q(6)	p(6)	Q(12)	p(12)
America	106	<0.001	168	<0.001
Europe	34.5	5.5e-06	51.9	6.4e-07
Japan	15.3	0.0183	31.7	0.0015
China	51.5	2.3e-09	83.2	1.0e-12
Asia Pacific / Other	95.9	<0.001	103	2.2e-16
Total	87	1.1e-16	130	<0.001

Series	Q(6)	p(6)	Q(12)	p(12)
China share of Asia Pacific	42.4	1.6e-07	68.6	5.7e-10

Test decision. For every series considered, the relevant p-values are below 0.05. The null hypothesis of no autocorrelation is rejected. Subsequent models must therefore incorporate explicit temporal dynamics rather than assume independent errors.

Table 6. Summary of hypothesis tests conducted in Phase 2

Test	H0	Result	Implication
ADF / PP on monthly Δlog	Unit root	H0 rejected for America, Europe, Japan, Asia Pacific, and Worldwide	Δlog is consistent with stationarity
KPSS on monthly Δlog	Stationarity	H0 not rejected for all five monthly series	Reinforces stationarity evidence
Ljung-Box	No autocorrelation through lag k	H0 rejected for the evaluated series	Temporal dependence must be modeled

4.4. Autocorrelation structure and seasonality

Table 7. Selected autocorrelation coefficients in the monthly series

Series	ACF(1)	ACF(12)	Interpretation
America	-0.3	0.63	Persistence remains; strong annual signal
Europe	-0.457	0.702	Strong annual seasonal component
Japan	-0.381	0.752	Strong annual seasonal component
Asia Pacific	-0.444	0.752	Strong annual seasonal component
Worldwide	-0.429	0.789	Strong annual seasonal component

For the Worldwide series, the ACF of first log differences is approximately -0.429 at lag 1, 0.766 at lag 3, 0.760 at lag 6, 0.789 at lag 12, and 0.666 at lag 24. The pattern combines very short-run reversal with persistent quarterly and annual signals.

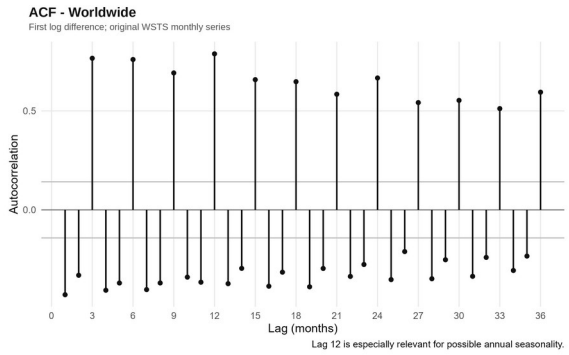


Figure 9a. Monthly ACF - Worldwide.

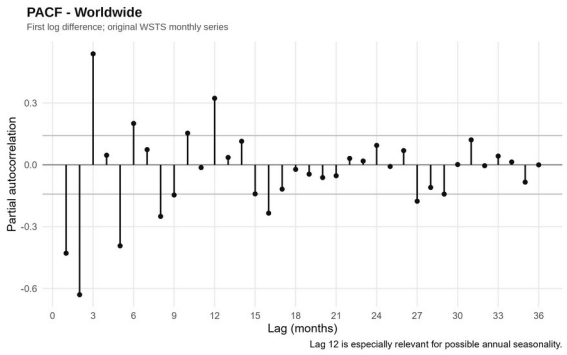


Figure 9b. Monthly PACF - Worldwide.

Source: author's calculations based on WSTS Historical Billings.

The final specification will need to account for this temporal structure. An OLS regression with independent errors would be inappropriate for inference on changes in level, trend, or policy interventions.

4.5. Monthly and smoothed series: effect of the moving average

The direct 2015-2025 comparison quantifies what the smoothing operation preserves and what it changes. Level correlations range from 0.965 to 0.986, indicating that the smoothed series reproduces long-run trajectories very closely.

Table 8. Volatility and level similarity: monthly versus smoothed series

Region	Monthly σ Δlog	Smoothed σ Δlog	Volatility reduction	Level correlation
America	0.11	0.0444	59.6%	0.986
Asia Pacific	0.0934	0.0274	70.7%	0.972
Europe	0.0826	0.0264	68.0%	0.966
Japan	0.0665	0.0213	68.0%	0.965
Worldwide	0.087	0.0262	69.9%	0.98

Table 9. Effect of smoothing on the autocorrelation structure

Region	ACF1 monthly	ACF1 smooth.	ACF3 monthly	ACF3 smooth.	ACF12 monthly	ACF12 smooth.
America	-0.268	0.727	0.596	0.05	0.629	0.405
Asia Pacific	-0.405	0.646	0.714	0.126	0.705	0.236
Europe	-0.435	0.32	0.378	-0.278	0.549	0.281
Japan	-0.348	0.564	0.596	-0.106	0.607	0.386
Worldwide	-0.387	0.707	0.744	0.108	0.746	0.376

The moving average reduces the volatility of first log differences by approximately 59.6% to 70.7%, depending on the region, and materially changes short-run autocorrelation. For Worldwide, for example, ACF(1) shifts from -0.387 in the monthly series to +0.707 in the smoothed series; ACF(3) falls from +0.744 to +0.108; and ACF(12) from +0.746 to +0.376.

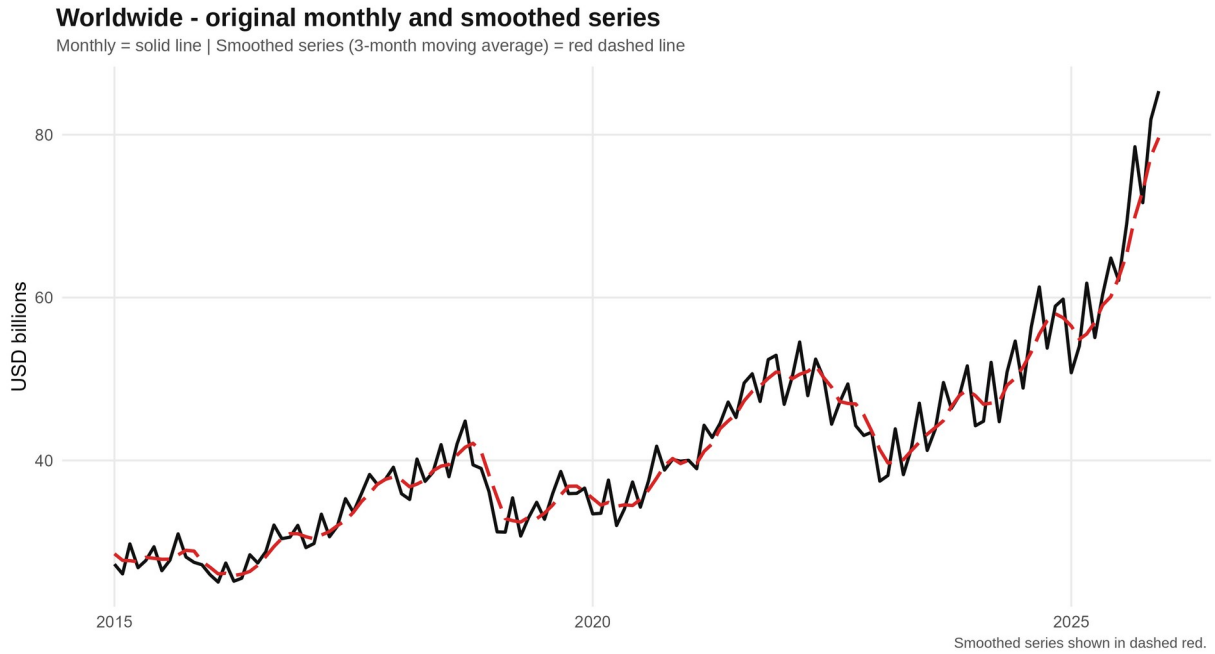


Figure 10. Worldwide: original monthly and smoothed levels, 2015-2025.
Source: author's calculations based on WSTS Historical Billings.

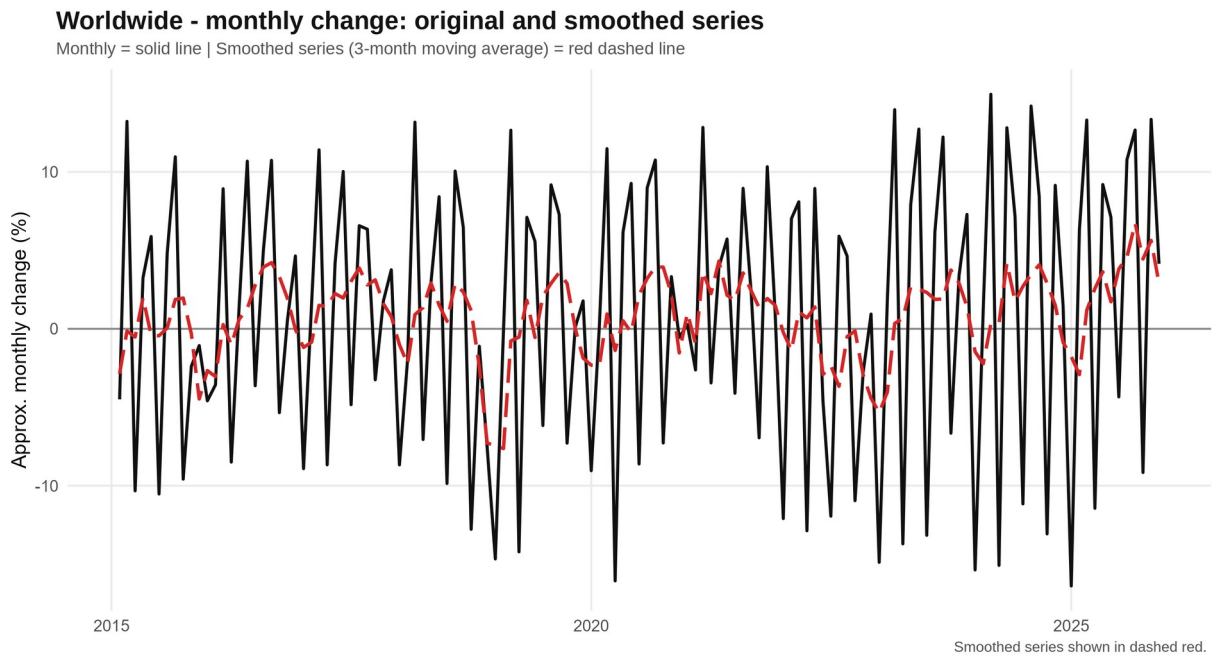


Figure 11. Worldwide: monthly change in the original and smoothed series.
Source: author's calculations based on WSTS Historical Billings.

The resulting methodological decision is to use the original WSTS monthly data as the main basis for temporal dynamics, seasonality, and structural-break analysis. The SIA/WSTS smoothed series is retained as the necessary extension when the research question requires China to be separated from the rest of Asia Pacific.

5. Preliminary conclusions and transition to Phase 3

Phases 1 and 2 establish a reproducible empirical foundation and show that the global semiconductor market between 2010 and 2025 combines strong long-run growth, sectoral cycles, seasonality, and meaningful regional rebalancing.

- America records the largest relative increase among the major regions and gains worldwide market share.
- Europe and Japan lose relative weight over the sample period.
- China grows strongly in absolute terms, but its share within Asia Pacific reaches a maximum in 2018 and subsequently declines.
- First log differences provide a statistically more suitable representation for part of the dynamic analysis than level series.
- Temporal dependence persists after differencing and must be modeled explicitly.
- The three-month moving average is useful for studying trends and comparing levels, but it does not substitute for original monthly data when the objective is to study short-run dynamics.

5.1. Next stage: Phase 3

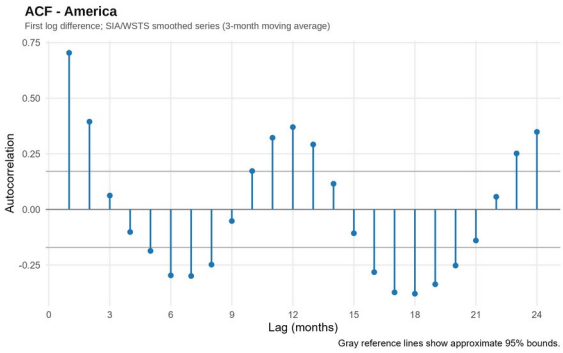
Phase 3 will begin with data-driven structural-break detection, particularly using Bai-Perron procedures, and with comparison between estimated breaks and the historical policy chronology. Results will also be framed as hypothesis tests: parameter stability against one or more structural breaks and, in intervention models, H0: beta_policy = 0 against H1: beta_policy != 0. Segmented or intervention specifications will then be evaluated with an error structure consistent with the autocorrelation and seasonality documented in Phase 2.

5.2. Interpretive limit

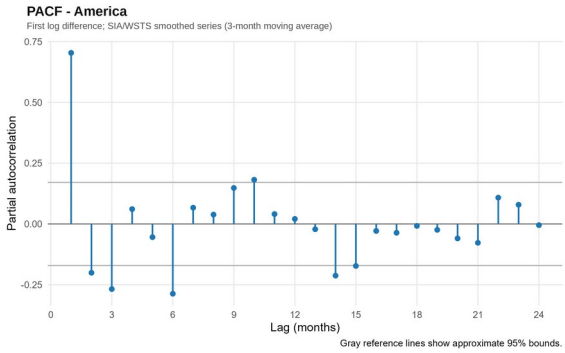
No result in this report demonstrates that the trade war, export controls, subsidies, or industrial policy caused the observed changes. The contribution of these two phases is to establish which patterns exist, how the series behave statistically, and which conditions a subsequent causal analysis must satisfy.

Appendix A. Diagnostics for the smoothed series

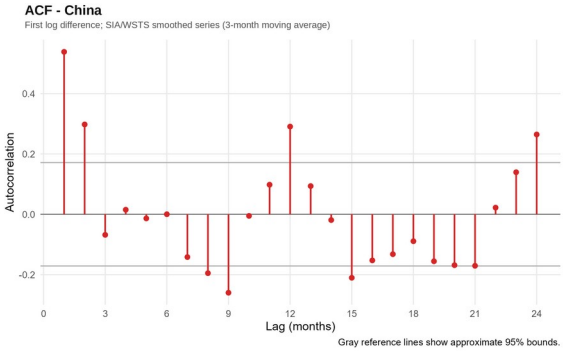
Autocorrelation and partial autocorrelation functions for the main SIA/WSTS smoothed series.



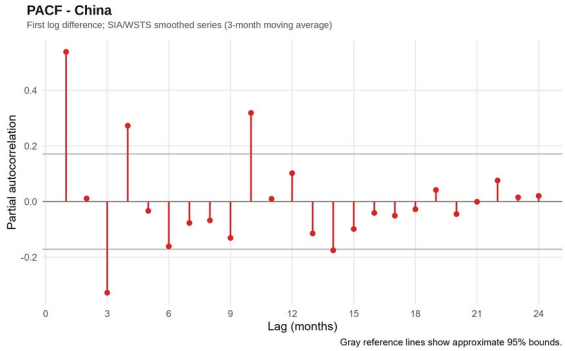
ACF - America.



PACF - America.

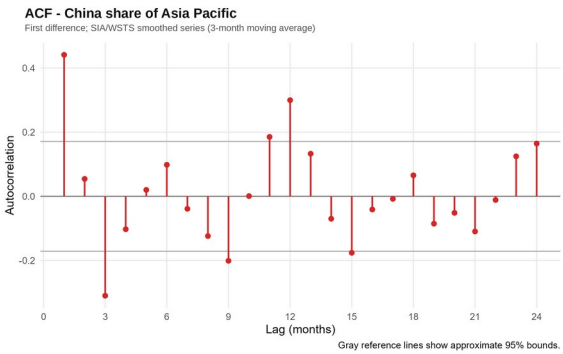


ACF - China.

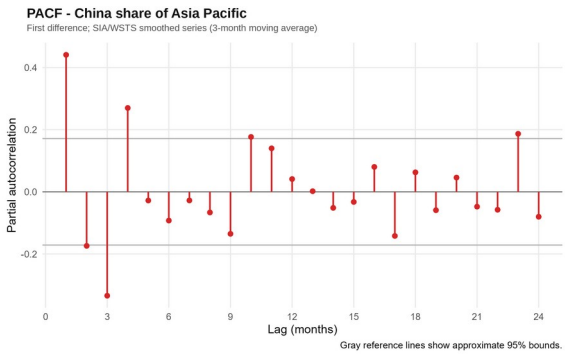


PACF - China.

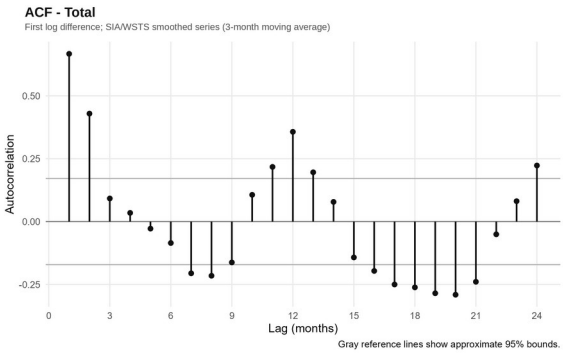
Source: author's calculations based on SIA/WSTS monthly market reports.



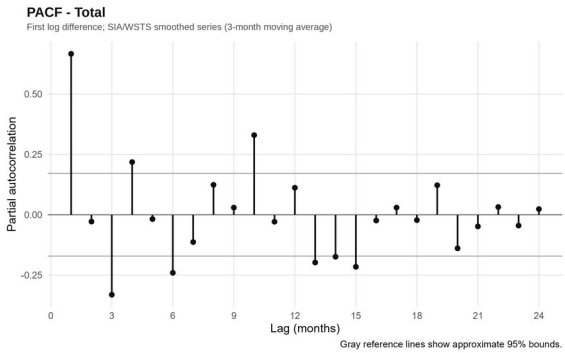
ACF - China share of Asia Pacific.



PACF - China share of Asia Pacific.



ACF - Total.

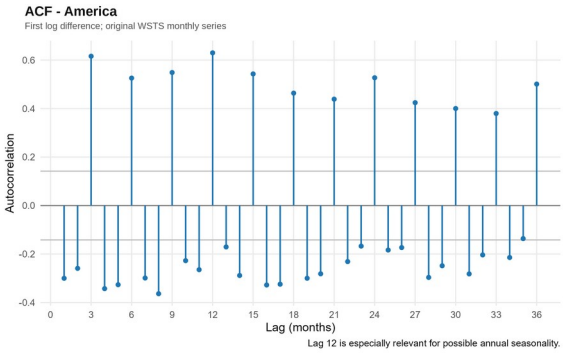


PACF - Total.

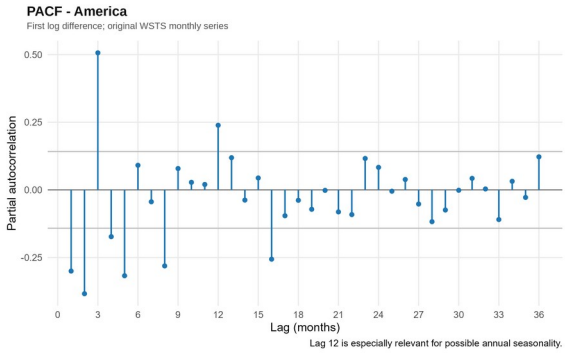
Source: author's calculations based on SIA/WSTS monthly market reports.

Appendix B. Diagnostics for the monthly data

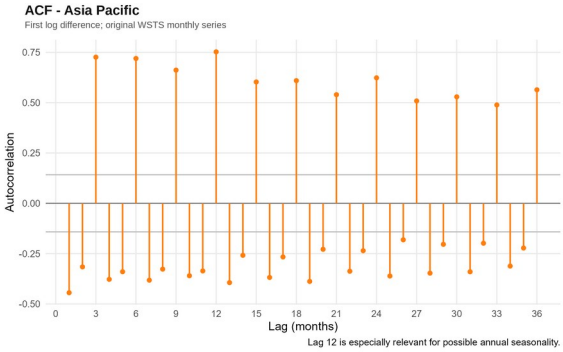
Autocorrelation and partial autocorrelation functions for first log differences in the WSTS monthly dataset.



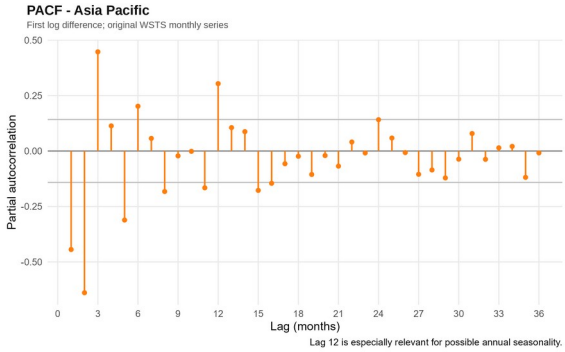
ACF - America.



PACF - America.

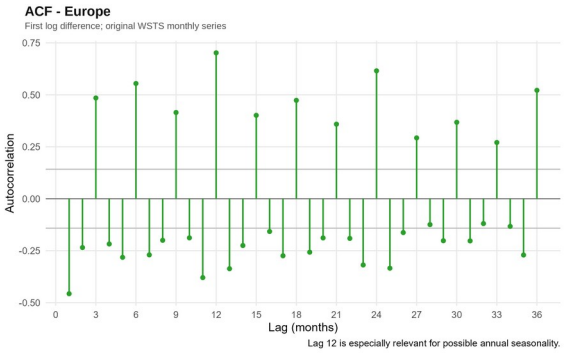


ACF - Asia Pacific.

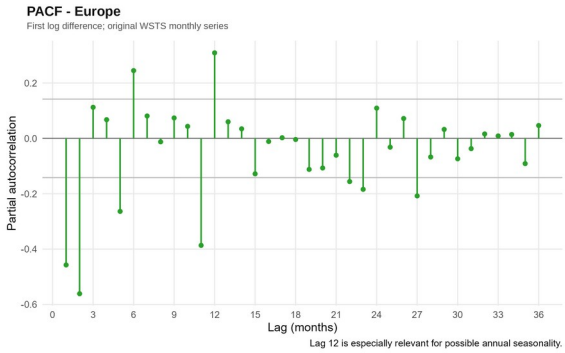


PACF - Asia Pacific.

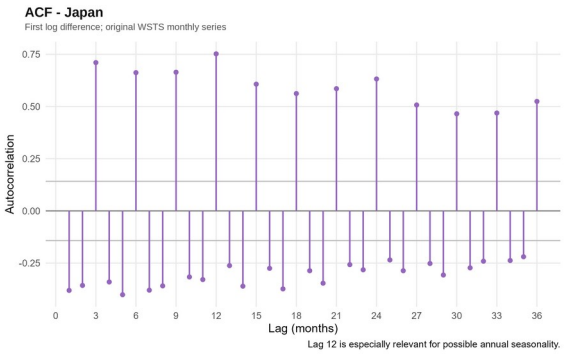
Source: author's calculations based on WSTS Historical Billings.



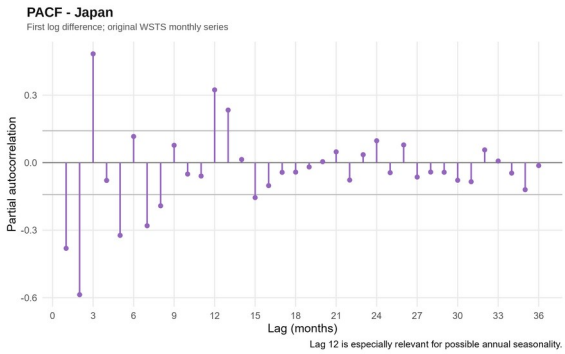
ACF - Europe.



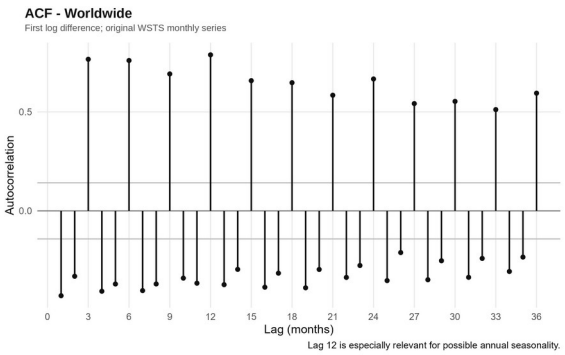
PACF - Europe.



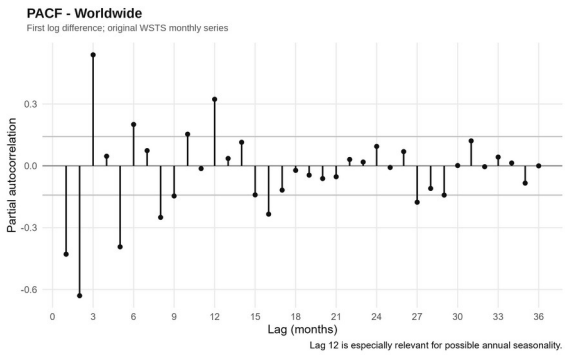
ACF - Japan.



PACF - Japan.



ACF - Worldwide.

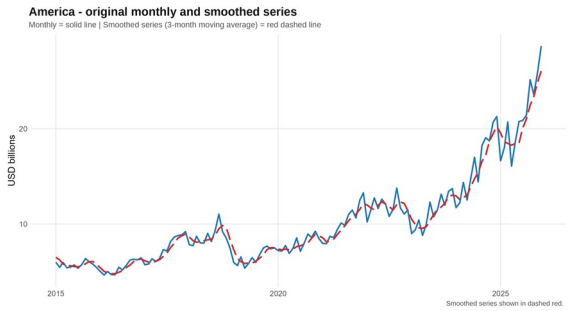


PACF - Worldwide.

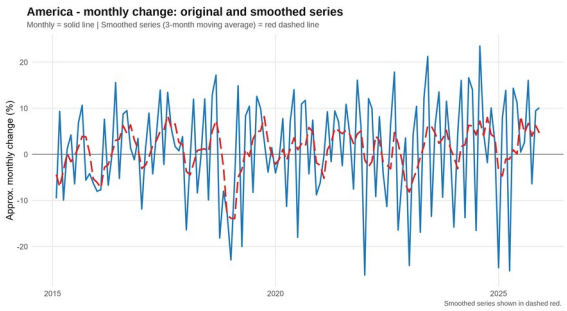
Source: author's calculations based on WSTS Historical Billings.

Appendix C. Additional comparisons

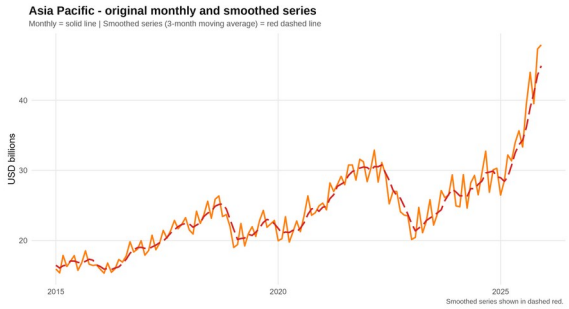
Comparisons between original monthly series and smoothed series for America and Asia Pacific.



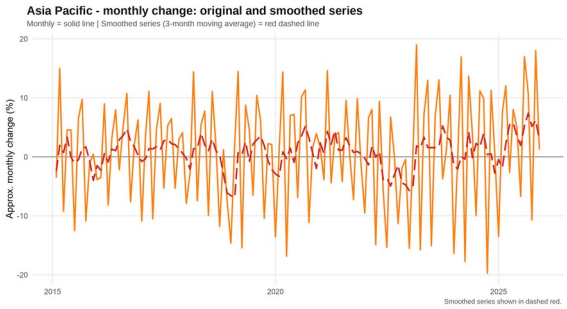
Levels - America.



Monthly change - America.



Levels - Asia Pacific.



Monthly change - Asia Pacific.

Source: author's calculations based on WSTS Historical Billings.

Sources and conventions

- WSTS Historical Billings: historical monthly series and three-month moving averages for America, Europe, Japan, Asia Pacific, and Worldwide.
- SIA Monthly Market Reports / WSTS: monthly reports used to reconstruct the China and Asia Pacific / Other decomposition.
- Observations in the smoothed series remain monthly; each data point summarizes three consecutive months and overlaps with adjacent periods.
- Monetary values are expressed in billions of US dollars.

This document is a reproducible interim output of the project and will be updated as structural-break tests, intervention models, and causal robustness exercises are incorporated.